

List of LaTeX environments

A [LaTeX environment](#) is one of the following:

Text inside a float environment is "floated" according to its placement, an optional parameter. The standard [report](#) and [article](#) classes use the default placement [tbp]. The float environments are figure and [table](#).

```
\begin{figure}[placement]
  % body of the figure
  \caption{figure title}
\end{figure}
```

Placement is one of the following

h (Here) - at the position in the text where the table environment appears.

t (Top) - at the top of a text page.

b (Bottom) - at the bottom of a text page.

p (Page) of floats - on a separate float page, which is a page containing no text, only floats.

The table environment has the same characteristics as the figure environment. [\[improvement needed: elaborate\]](#)

The **list** environments are [description](#), enumerate, itemize, and the generic [list](#), which is seldom used in text, but often in macros. Within a list environment item commands are used to identify the items in the list.

The **description** environment produces a labeled list.

```
\begin{description}
\item [label] First item
\item [label] Second item
...
\end{description}
```

The **enumerate** environment produces a numbered list. At least one item is needed. Enumerate lists can be nested inside other enumerate lists, up to four levels deep.

```
\begin{enumerate}
\item First item
\item Second item
```

...

`\end{enumerate}`

The **itemize** environment creates an unnumbered, or "bulleted" list.

`\begin{itemize}``\item First item``\item Second item`

...

`\end{itemize}``\begin{list}{label}{fmt-params}``\item First item``\item Second item`

...

`\end{list}`

label

specifies how items should be labeled. This argument is a piece of text that is inserted in a box to form the label. This argument can and usually does contain other LaTeX commands.

fmt-params

contains commands to change the spacing parameters for the list. An empty argument will select all default spacing which should suffice for most cases.

The format parameters are: (See external reference for a diagram of these)

`\itemsep``\labelsep``\labelwidth``\leftmargin``\listparindent``\parsep``\parskip``\partopsep``\rightmargin``\topsep`

The math environments are [math](#), [displaymath](#), [array](#), [eqnarray](#), [equation](#), and [theorem](#)

The math environment is used within [paragraph mode](#) to place mathematical symbols inline with the paragraph's text. The following are all equivalent:

`\(... \)``$ ... $``\begin{math}...\end{math}`

In a wiki, the `<math> ... </math>` tags create a math environment.

The following are equivalent:

```
\[ ... \]

$$...
\begin{displaymath}...\end{displaymath}$$

```

Note: The AMS Short Math Guide recommends against using $...$ to enter displaymath mode, because it is not documented in the TeX manual, and because it interferes with the proper operation of the fleqn option.

Main article: [array \(LaTeX environment\)](#)

The array environment is used to make a table of information, with column alignment (left, center, or right) and optional vertical lines separating the columns.

```
\begin{array}{col1col2...coln}
item11 & item12 ... & item1n\\
item21 & item22 ... & item2n\\
...
itemn1 & itemn2 ... & itemnn\\
\end{array}
```

Each of col1, col2, etc. is a single letter,

c - for centred
l - for flush left
r - for flush right

The column format letters can be interspersed with vertical bars (|), which result in rendering vertical bars as needed in the formatted matrix.

If a horizontal line is needed in the matrix, [\hline](#) can be used.

Within the matrix itself, columns are separated by an &, and each row ends with a double backslash, \\.

```
\begin{eqnarray}
math formula 1
math formula 2
...
\end{eqnarray}
```

The **eqnarray** environment is

An equation number is placed on every line unless that line has a \nonumber command. The command \lefteqn is used for splitting long formulas across lines. It typesets its argument in

display style flush left in a box of zero width.

The **equation** environment centers the equation, and places an equation number in the right margin.

```
\begin{equation}
... math formula
\end{equation}
```

An asterisk is used for unnumbered equations:

```
\begin{equation*}
... math formula
\end{equation*}
```

An equation can be given a label using the `\label` command, which can later be referenced using the `\eqref` command.

```
\begin{equation}\label{xyz}
... math formula
\end{equation}
```

Later, the equation can be referenced, as an example, by
... using equation `\eqref{xyz}`, one can derive the following...

`\eqref{xyz}` generates a parenthesized equation number; it is equivalent to `(\ref{xyz})`. The preamble command, `\numberwithin{equation}{section}` can be used to number equations with the two-level name m.n (section number.equation number).

The **subequations** environment affixes letters a, b, c... to the end of the equation number within the environment.

```
\begin{equation}\label{first}
x=y+z
\end{equation}
some intervening text
\begin{subequations}\label{grp}
\begin{align}
a&=b+c\label{second}\\
d&=e+f+g\label{third}\\
h&=i+j\label{fourth}
\end{align}
\end{subequations}
```

numbers equation $x=y+z$ as, for example, 2.9, then the group $a=b+c$, $d=e+f+g$, and $h=i+j$ will be numbered, respectively, 2.10a, 2.10b, and 2.10c. The "parent" equation number, 2.10, is assigned to the label "grp".

The **multline** environment lets you split equations over multiple lines, aligning the beginning of the equation with the left margin, and the end of it with the right margin. Equations are numbered.

```
\begin{multline}
a+b+c+d+e+f\\
\end{multline}
```

```
+i+j+k+l+m+n
\end{multline}
```

The `gather` environment gathers and centers equations; each equation is numbered; not available in wiki.

```
\begin{gather} a_1=b_1+c_1\\
a_2=b_2+c_2-d_2+e_2
\end{gather}
```

The `align` environment lets you use the ampersand (&) to align multiple equations; each equation is numbered; not available in wiki.

```
\begin{align}
a_{11}&=b_{11}&
a_{12}&=b_{12}\\
a_{21}&=b_{21}&
a_{22}&=b_{22}+c_{22} \end{align}
```

The `flalign` environment lets you use the ampersand (&) to align multiple equations. The leftmost and rightmost columns are aligned with the margins.

```
\begin{flalign*}
a_{11}&=b_{11}&
a_{12}&=b_{12}\\
a_{21}&=b_{21}&
a_{22}&=b_{22}+c_{22} \end{flalign*}
```

The `theorem` environment produces "Therem x" in bold face, followed by your theorem text. [\[improvement needed: elaborate - how is theorem name specified?\]](#)

```
\begin{theorem}
... theorem text
\end{theorem}
```

Main article: [matrix \(LaTeX environment\)](#)

The matrix environments are `matrix`, `pmatrix`, `bmatrix`, `Bmatrix`, `vmatrix`, `Vmatrix`, and `smallmatrix`. Each provides a table for expressions, aligned in rows and columns. The main difference between the various types of matrix is the kind of delimiters that surround them.

Each row of a matrix ends with two backslashes (`\\`). Each column ends with an ampersand (&).

The `cases` environment is used to left-brace a group of equations or a piecewise-defined function, for example,



The `align` environment can be used to align equations.



The `alignat` environment can be used to align equations, and explicitly specify the number of "equation" columns. An equation column has two parts, separated by the equals-sign. Essentially, this is an array with alternating right-aligned and left-aligned columns. The required parameter of `alignat` is the maximum number of ampersands in a row plus 1, and then divided by 2. One use of `alignat` is to explicitly specify the amount of horizontal space between columns by including the required spacing in the first row.



For some reason, in the wikia wiki, the "align" and "alignat" environments cause errors.

[[improvement needed](#): elaborate]

The **paragraph** environments are [center](#), [flushleft](#), [flushright](#), [minipage](#), [quotation](#), [quote](#), [verbatim](#), and [verse](#). Within a paragraph environment, each line must be terminated with a double backslash, `\\`.

Related commands are [centering](#), [raggedright](#), [raggedleft](#), [parbox](#), [footnote](#), [footnotetext](#), and [verb](#)

The `center` environment allows you to create a paragraph consisting of lines that are centered within the left and right margins on the current page.

The `flushleft` environment allows you to create a paragraph consisting of lines that are flushed left, to the left-hand margin.

The `flushright` environment allows you to create a paragraph consisting of lines that are flushed right, to the right-hand margin.

The **minipage** environment is similar to the `\parbox` command. It is not advisable to nest `minipage` environments, because it messes up footnote positioning.

```
\begin{minipage}[position]{width}
... text
\end{minipage}
```

The margins of the `quotation` environment are indented on the left and the right. The text is justified at both margins and there is paragraph indentation.

The margins of the `quotation` environment are indented on the left and the right. The text is justified at both margins without paragraph indentation

Uses typewriter (`\tt`) style, similar to the `\verb` command.

The verse environment is designed for poetry. The margins are indented on the left and the right. End each lines with `\\` and use a blank line to separate the stanzas.

```
\begin{picture}(width,height)(x offset,y offset)
```

```
... picture commands
```

```
\end{picture}
```

picture commands are

```
\circle
```

```
\dashbox
```

```
\frame
```

```
\framebox
```

```
\line
```

```
\linethickness
```

```
\makebox
```

```
\multiput
```

```
\oval
```

```
\put
```

```
\shortstack
```

```
\vector
```

[improvement needed: elaborate]

In the tabbing environment,

- Tab stops are set with `\=`.
- `\>` advances to the next tab stop.
- `\+` ends a line, and begins the next line one tab stop to the right.
- `\-` ends a line, and begins the next line one tab stop to the left.

[improvement needed: elaborate]

```
\begin{tabbing} function \= fact(n : integer) : integer;
```

```
\> begin \= \+
```

```
\> if \= n > 1 then \+
```

```
fact := n * fact(n-1) \-
else \+
```

```
fact := 1; \-\-
end;
```

```
\end{tabbing}
```

[improvement needed: formatting]

Related commands are

The `tabular` environment uses ampersands (&) to separate columns.

```
\begin{tabular}[pos]{cols}
column 1 entry & column 2 entry ... & column n entry
```

```
...
```

```
\end{tabular}
```

or

```
\begin{tabular*}{width}[pos]{cols}
column 1 entry & column 2 entry ... & column n entry
```

```
...
```

```
\end{tabular*}
```

[improvement needed: elaborate]

The `thebibliography` environment is used to print a bibliography. Related commands are `bibitem`, `cite`, `nocite`

```
\begin{thebibliography}{widest-label}
\bibitem[label]{cite_key}
```

```
...
```

```
\end{thebibliography}
```

The `titlepage` environment is used to print a title page. It has no printed page number or page heading. The following page is numbered page 1. A related commands is `maketitle`.

```
\begin{titlepage}
```

```
... title page text
```

```
\end{titlepage}
```

